



Evaluation of critical heat flux of ATF candidate coating materials in pool boiling

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ARTICLE INFO

Keywords:

Critical heat flux
Accident tolerant fuel cladding
Non-hydrodynamic effect
Thermal activity
Thermal spreading ability

ABSTRACT

The critical heat flux (CHF) of potential accident tolerant fuel (ATF) cladding materials and concepts, Cr and FeCrAl coatings on zirconium-alloy substrates, was investigated experimentally. A cold spray process, which involves the propulsion of high velocity powder particles on to the substrate surface to form a coating upon impact, was used to create the coatings. Surface parameters such as contact angle, surface roughness, and morphology that are associated with CHF in pool boiling were characterized. Non-hydrodynamic effects of newly developed ATF candidate coating materials on CHF were also evaluated. The effects of thermal properties and coating thickness and expected challenges for evaluation of CHF for new ATF cladding materials development are discussed.

1. Introduction

All current Light Water Reactor (LWR) nuclear power plants are designed to cope with a range of anticipated operational occurrences (AOO) as well as a comprehensive set of design basis accidents (DBAs). For either an AOO or a DBA, the owner/operator is required to demonstrate that the plant is designed with safety systems to cope with such events and assure that adequate cooling is provided to the fuel/cladding system in the reactor core so as to remain within certain design criteria and preclude fuel failure. These fuel design criteria are set to establish a limit on the allowable peak clad temperature as well as to establish a limit on the amount of clad oxidation, thereby maintaining the fuel integrity. Such criteria also limit the metal-water chemical reactions that can produce combustible hydrogen. After the events at Fukushima in 2011, the focus has shifted to the important issue of developing strategies to increase the capability of LWRs to passively respond to accidents while maintaining long-term cooling for extended time periods (Farmer et al., 2016; Yang, 2014). The nuclear reactor community is considering several approaches to achieve this goal. One approach is to develop accident tolerant fuel (ATF) designs that have superior oxidation resistance at high temperatures (Carmack et al., 2013). The improved oxidation resistance of the new nuclear fuels is expected to reduce combustible hydrogen generation even under high temperature accident conditions, thereby providing increased coping times for compensatory actions to mitigate system damage (Ott et al.,

2014; Robb et al., 2016, 2017). In addition, the development of ATF concepts could potentially improve fuel efficiency during normal operating conditions owing to their superior characteristics (Bragg-Sitton, 2014).

Initial research on ATF development has been focused largely on development of materials and manufacturing processes, but evaluating thermal hydraulic characteristic these advanced cladding material concepts is also an important consideration towards the finalization of the ATF concepts. Critical heat flux (CHF) represents the boiling transition point from nucleate boiling regime that has high heat transfer to low heat transfer film boiling regime. Due to the significant heat transfer at CHF, surface temperature abruptly increase, which would threaten the heating integrity of the materials' surface. Therefore, CHF measurements of ATF cladding concepts must be evaluated as an important parameter for determining safety margin of the new ATF systems (Ali et al., 2018; Seo et al., 2016; Son et al., 2019).

CHF is a function of operating conditions (e.g., pressure, subcooling, mass flow rate, and vapor quality), system geometry, and surface conditions. Recent research has focused on surface effects of the heated wall surface on boiling transition points including CHF. In particular, wettability, roughness, and thermal properties have been considered as main surface characteristics that could affect CHF (Ahn et al., 2011; Jo et al., 2015; Kandlikar, 2001; Kim et al., 2010; Liter and Kaviany, 2001). In this regard, the surface characteristics of the newly developed ATF cladding concepts must be evaluated in conjunction with

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<https://doi.org/10.1016/j.nucengdes.2019.110166>

Received 29 May 2019; Received in revised form 17 June 2019; Accepted 18 June 2019

Available online 22 June 2019

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measurements of CHF of the ATF material, such that together this information can be used for the safety-related licensing process. It has been noted that CHF values for same material can vary depending on cladding and coating fabrication methods. Son et al. (2018) prepared three different Cr coating surfaces on Nichrome wire with nanofluid deposition and two different sputtering methods, and found significant difference in CHF between the three types of Cr-coatings. Radio Frequency (RF) sputtered Cr coating showed 1.9 times enhancement in CHF compared to non-coated surface but Direct Current (DC) sputtered Cr coating showed almost no enhancement in CHF. Kam et al. (2015) performed pool boiling experiments with flat SiC physical vapor deposition (PVD) sputter coated samples and electroplated Cr coated samples. Interestingly, the electroplating Cr coated surface on stainless steel plate showed lower heat flux than bare surface including non-coated stainless steel and Zircaloy-4 plates, which is opposite to the results reported by Son et al. (2018). On the other hand, another DC sputtered Cr coatings reported in recent showed almost no change in CHF between with and without the coating even different intrinsic surface roughness (Son et al., 2019). The study concludes that CHF values for coated surfaces should be confirmed experimentally even if the heating surfaces are made of a same material but prepared with different coating methodologies or parameters.

In this current work, an industry-scale cold spray coating technology was used to produce a thin Cr coating on zirconium-alloy flats. This approach is recognized as one of the ATF cladding concepts being considered by industry. Surface features of the cold spray Cr coated samples were evaluated, and key considerations to analyze this new ATF concept for CHF are discussed.

2. ATF candidate material preparation and surface characterization

2.1. Sample preparation

A commercial Cold Gas Technologies (CGT) Kinetiks 4000/34 unit system at the University of Wisconsin, Madison was used to produce of dense Cr coatings and FeCrAl coatings. Details of the cold spray processes for the coatings can be found in previous studies (Maier et al., 2019; Yeom et al., 2018). All coating parameters including particle sizes are optimized in previous studies for ATF Zr-alloy cladding (Maier et al., 2019; Yeom et al., 2018). Since the purpose of this study is evaluating CHF with actual nuclear cladding surfaces, we focus on the best coating process for ATF cladding conditions. ZIRLO sheets with thickness of 0.8 mm, provided by Westinghouse Electric Company, were used as the substrate for the coatings and untreated ZIRLO clad materials also served as the base-line reference. The as-deposited coatings were polished with 600 grit SiC abrasive paper to reduce surface irregularities and to bring the coating thickness down to approximately from 64 μm to 79 μm for Cr coating and about 40 μm for FeCrAl, resulting in the surface finish representative of commercial cladding in LWRs. The same surface finish was applied to bare ZIRLO sheet, used to generate baseline CHF data for comparison with the coated samples. All the Cr coated ZIRLO sheet, FeCrAl coated ZIRLO sheet, and uncoated ZIRLO sheet were sectioned into small coupons

(20 mm $\times$ 20 mm $\times$ 0.8 mm thickness), which were used as the heated surface in pool boiling experiment. Fig. 1 shows the photographs of the progression in sample preparation for as-cold spray condition to surface condition for CHF measurements.

2.2. Surface characterization

Wettability represents interaction force between working fluid/gas and heating surfaces, so it dictates the contact angle of water with heating surfaces which then affects bubble interface dynamics. Strong adhesion between liquid and heating surfaces results in low contact angle, and is referred to as hydrophilic surface, while a high contact angle signifies a hydrophobic surface. The hydrophilic surface is known to have high CHF with low boiling heat transfer, while the hydrophobic surface shows lower CHF as demonstrated in earlier experiments (Jo et al., 2011, 2014; Kandlikar, 2001). Kandlikar demonstrated CHF enhancement with decreasing contact angle by analyzing the force balance between the evaporation momentum inertia and the surface tension force at the boundary of bubble interface on the heating surface. Later, Ahn et al. (2011) proposed a correction to the Kandlikar's model by introducing a constant to fit their experiment data obtained from small circular heaters.

The surface structure could also affects liquid supplied to the dry-out area at CHF. Recent literature has shown that physically structured surfaces can lead to improvements in CHF up to 300% compared to a plain surface (Kim et al., 2010, 2015). The physical structures of a heating surface can interrupt bubble growth and the resistance provided by these structures to bubble dynamics increases as surface roughness increases. The increased resistance due to surface roughness can be represented by the ratio of the total area to the projected area, which denotes increased surface area by surface topographical elevations compared to an ideally smooth surface. Also, capillary force by micro/nano-surface structures have been considered along with their surface morphological characteristics. The surface structure would generate frictional force and also supply water by capillary. In addition, surface structures could affect hydrodynamic instability by influencing liquid and vapor paths. The cumulative effect of these surface roughness effects could be quantitatively correlated to bubble dynamics and liquid spreading ability in structures, which is closely related to the rewetting of a dry patch during boiling and CHF (Ahn et al., 2012; Kim et al., 2015).

In this regard, the contact angle and surface structure of the heating surfaces have been carefully characterized. Table 1 shows measured roughness parameters, and Fig. 2 shows SEM images of sample surfaces prepared for this study. Measured contact angles are shown in Fig. 3. Since we prepared the heating surfaces to be very similar to commercial-level cladding surface, there is no significant difference of the surface parameters between the coated surfaces and the uncoated surfaces. In particular, the surface roughness is almost same for all surfaces without capillary wicking. The differences in wettability could stem only from differences in chemical potential of the materials' surfaces but there is typically no significant difference between metallic alloys in this regard. Thickness of the coatings is in range of 40–80 μm based on multiple micrometer measurements.

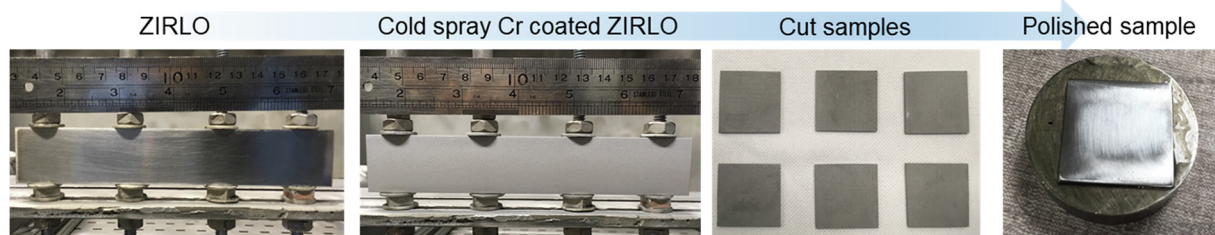


Fig. 1. Sample preparation procedure for cold spray coated samples used in this study.

Table 1

Summary of roughness parameters: Ra (arithmetic average of the roughness), PV (peak-to-valley distance), RMS (root mean square of the microscopic peaks and valleys).

Parameter	ZIRLO	Cr coated ZIRLO	FeCrAl coated ZIRLO	SS304
Ra (μm)	0.12	0.074	0.067	0.07
PV (μm)	3.1	3.1	2.9	2.5
RMS (μm)	0.17	0.12	0.10	0.11

3. Pool boiling facility and experimental results

3.1. Pool boiling facility

Pool boiling experiments were conducted with the coated ATF cladding materials and ZIRLO surfaces under saturated conditions using the experimental facility shown in Fig. 4. Pool boiling facility consists of a water pool, an immersion heater with a PID controller and a thermocouple, a heater block, and a sample holder. A rectangular pool ($14 \times 14 \times 14 \text{ cm}^3$) constructed with aluminum is used for this experiment. The immersion heater was installed on a window wall of the boiling pool to maintain working fluid (water) temperature in saturated condition with the water level above a certain minimum level during the experiments. The copper heating block has four cartridge heaters to generate heat and three thermocouples to measure transferred heat with temperature difference at various given positions. Three thermocouple holes were drilled along a line on the heating block to quantify 1D transferred heat flux. Using ANSYS FLUENT, it was confirmed that the heater block geometry had uniform temperature distribution on the cross-section at the thermocouple hole locations, which is necessary to assume 1D heat flux transfer in the heater block. The sample holder is made of Polyether ether ketone (PEEK) thermoplastic polymer. Due to

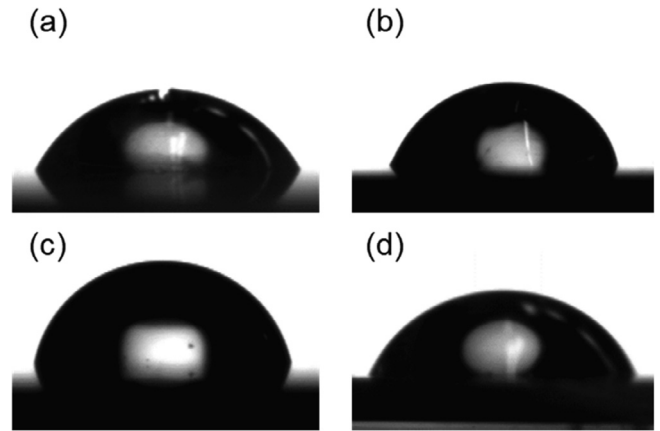


Fig. 3. Contact angles of surfaces with water droplets: (a) ZIRLO – $64^\circ \pm 3^\circ$, (b) Cr coated ZIRLO – $77^\circ \pm 6^\circ$, (c) FeCrAl coated ZIRLO – $89^\circ \pm 9^\circ$, and (d) 304 stainless steel – $68^\circ \pm 3^\circ$.

the low thermal conductivity of PEEK, heat loss from the sample to the sample holder could be ignored. The sample specimen size was $20 \times 20 \text{ mm}$ but the heating block was $18 \times 18 \text{ mm}$, providing for an air gap between actual heating area and the wall of the sample holder that minimizes heat loss.

Before turning on the heater block, all experiments conducted degassing procedure with the immersion heater by maintaining saturated temperature of the working fluid for at least 1 hr. It removed dissolved gas from the working fluid, which can affect boiling performance. After degassing procedure, electrical load of the cartridge heaters were controlled to heat up the boiling sample. The power level was step-wisely increased to avoid rapid transition in boiling. All power steps were

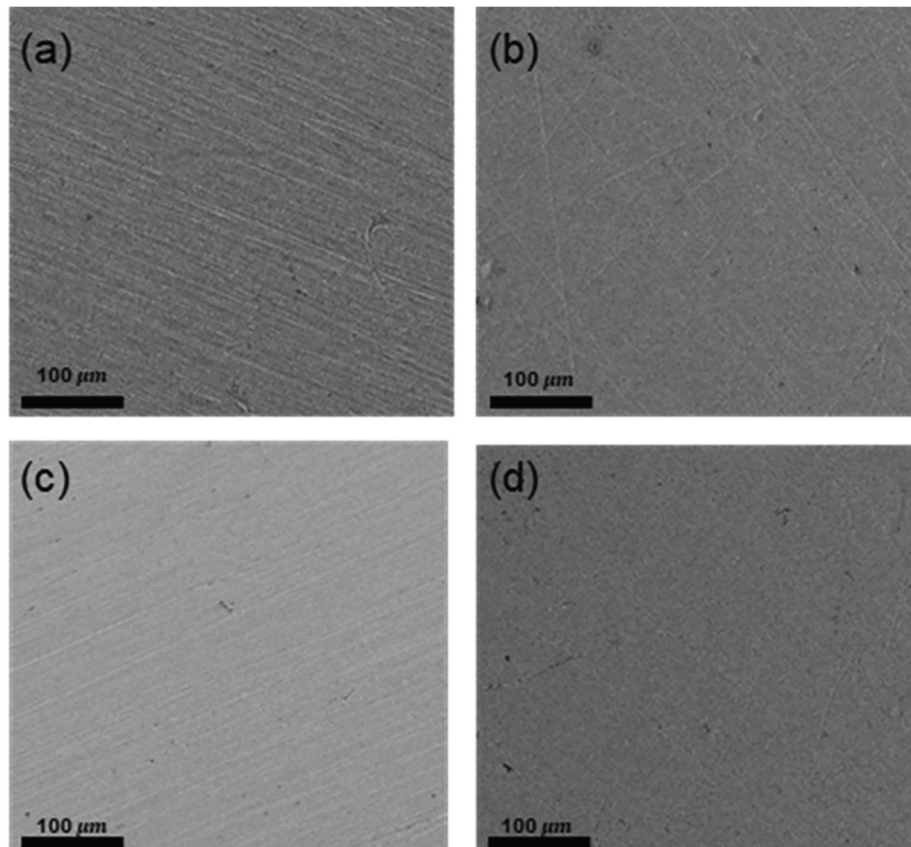


Fig. 2. SEM images of the surfaces of the samples prepared for this study: (a) ZIRLO, (b) Cr coated ZIRLO, (c) FeCrAl coated ZIRLO, and (d) 304 stainless steel.

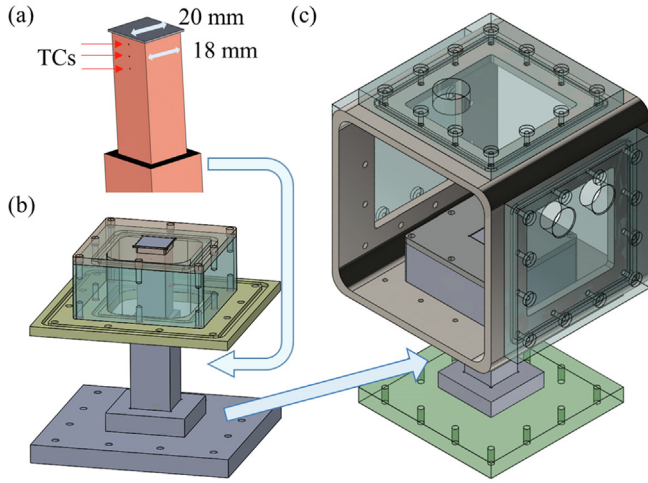


Fig. 4. Illustrations of the pool boiling facility designed and constructed for this study: (a) heater block and sample specimen, (b) assembly of sample holder and heater block part, and (c) full assembly of the facility.

determined based on experimental confirmation that there was no change by transition phenomena in the data value when the power level was raised slightly lower and slower. CHF values are defined by the top two thermocouples when the thermocouple temperature abruptly increases with the measured heat flux decrease.

The uncertainty analysis for measured heat flux in this facility was conducted based on work of previous researchers (Coleman and Steele, 1999). In this study, based on 1D heat conduction, the transferred heat flux is calculated as,

$$q'' = k \frac{\partial T}{\partial x} = k \frac{T_2 - T_1}{x_2 - x_1} \quad (1)$$

q'' is heat flux, k indicates thermal conductivity of the heater block, T_1 and T_2 are measured temperature from the heater block location x_1 and x_2 , respectively. The measured heat flux is a function of measured temperatures and the hole locations in the heater block. So, the uncertainty expression for the heat flux becomes

$$U_{q''}^2 = \left(\frac{\partial q''}{\partial T} \right)^2 U_T^2 + \left(\frac{\partial q''}{\partial x} \right)^2 U_x^2 \quad (2)$$

where $U_{q''}$, U_T , and U_x represent the uncertainties in q'' , T , and x , respectively. The general uncertainty expression is

$$\frac{U_{q''}^2}{q''^2} = \left(\frac{T}{q''} \frac{\partial q''}{\partial T} \right)^2 \frac{U_T^2}{T^2} + \left(\frac{x}{q''} \frac{\partial q''}{\partial x} \right)^2 \frac{U_x^2}{x^2} \quad (3)$$

With the uncertainty magnification factors (in parentheses) equals 1.0, it would be simplified as

$$\frac{U_{q''}^2}{q''^2} = \frac{U_T^2}{T^2} + \frac{U_x^2}{x^2} \quad (4)$$

The thermocouple uncertainty is assumed to be 1 °C as stipulated by commercial vendor (K-type thermocouple, OMEGA engineering). The measured temperatures from the heater block at CHF ranged from 200 °C to 300 °C. The machined hole locations were calibrated with image analysis, allowing us to estimate the uncertainty of the hole locations to be 0.5 mm. Accounting for instrument errors, the maximum uncertainty of heat flux is estimated to be less than 1%.

3.2. Pool boiling experimental results

In this study, four different surface materials (ZIRLO, Cr-coated ZIRLO, FeCrAl-coated ZIRLO, and stainless steel 304) are used as heating surfaces in the pool boiling facility. Fig. 5 shows the

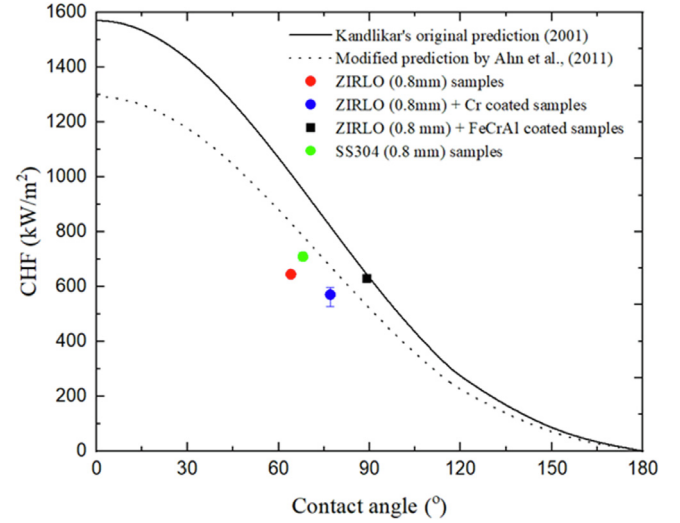


Fig. 5. CHF results as a function of contact angle on the surfaces obtained in this study and compared to previous models.

experimental result for these materials' surfaces. We reproduced at least three samples for each surface condition, and average CHF values are plotted with data range of the CHF values are plotted in the figure. As expected from the similar surface characteristics, all samples show similar CHF values. The values for ZIRLO, Cr coated and FeCrAl coated ZIRLO are 645 kW/m², 571 and 630 kW/m², respectively. The variation between the coated samples and the non-coated samples is less than 12%. The commercial material SS304 also shows similar CHF (709 kW/m², ~10% higher than ZIRLO). We believe that the small differences in wettability on different metal materials results in small change in CHF but the difference is not significant according to the experimental results. We compare the experimental results with prediction models having the contact angle effect on CHF. There are two prediction models in the Fig. 5; one is the original Kandlikar's model and the other one is the modified one with the correction factor for a circular heater with a diameter of 10 mm (Ahn et al., 2011). Although there is slight difference in the sample dimension between current study and Ahn's study, the modified Kandlikar model with the correction factor shows values comparable to our experimental results.

Fig. 6 shows the comparison with different Cr-coated surfaces performed as ATF coating design in previous researches. We plot the result in two ways to show actual values and relative change of CHF and contact angles. Depending on the coating technique, highly scattered trend is revealed. The three different Cr-coated surfaces prepared by nanofluid deposition, RF sputtering, and DC sputtering show low contact angles with additional capillary wicking (Son et al., 2018). The combination of wettability and surface structure resulted in improvement of CHF values compared to non-coated bare surface (Nichrome wire). Kam et al. (2015) also used Cr-coated surfaces using electroplating but the result contact angle of the Cr-coated surface was higher than non-coated bare surface of stainless steel and zircaloy-4. CHF values on the Cr-coated sample by electroplating were lower than not only the bare stainless steel surface but also Zircaloy-4. Very recent study also used DC sputtering to put Cr-coating on differently grinded stainless steel 316 tube to see the effects of intrinsic surface roughness and coating (Son et al., 2019). In the Fig. 6, Cr-coated 1 and 2 by Son et al. (2019) indicate different stainless steel substrates grinded with 800 and 60 grit sandpapers, respectively. They found the intrinsic roughness effect on CHF but Cr-coated samples did not affect CHF values. Compared to this, this cold spray coating that is being considered in nuclear industry to fabricate Cr-coated ZIRLO cladding shows no significant change in surface characteristics and CHF compared to bare ZIRLO surfaces, which is expected from identical surface characteristics between coating or non-

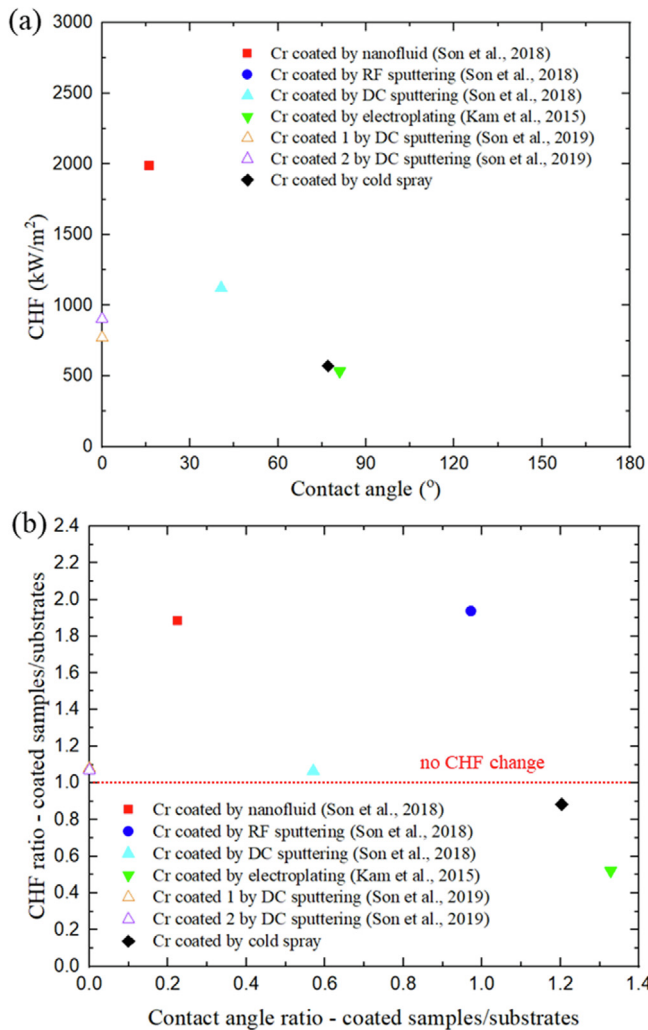


Fig. 6. Comparison of average CHF Cr coatings prepared by other approaches: (a) actual CHF and measured contact angles and (b) CHF ratio of Cr coated samples and non-coated samples as a function of contact angle ratio of the Cr coated samples and non-coated samples.

coating reference due to the similar surface finish for nuclear cladding surface. This result is significant in discussion of safety issues of ATF candidate reactor cladding materials.

3.3. Discussion: Thermal properties effect on CHF

In this study, we have focused on surface characteristics, which could affect hydrodynamics of boiling. However, we also need to consider non-hydrodynamic effects of the new ATF materials on CHF. Due to the already specified nuclear cladding geometry and operating condition, dominant non-hydrodynamic parameters of ATF cladding design are limited to thermal properties and the thickness of the coatings. The samples evaluated in this study are the coated cladding concepts which are being viewed as near-term strategies for ATF development, so only thin coating layers are fabricated (5–10% of substrate thickness) and the effect of thermal properties are assumed not to be significant. However, for long-term strategy of ATF development involves complete elimination and replacement of current zirconium-alloy cladding with new materials such as FeCrAl alloys or SiC-SiC_f composites, the discussion of the thermal properties and material thickness effects is necessary. Also, coating defects (e.g., impurity and porosity), which depend on coating deposition method, could have a significant effect in thermal properties. In this regard, we review previous studies and

discuss the thermal properties and the thickness effects on CHF.

Previously a number of experiments reported higher CHF on thicker samples than thinner samples, which led to a discussion on the effects and inter-relationship between thickness and thermal properties on CHF. CHF increased as the heating surface thickness increased, and then approached a certain asymptotic value but the threshold thicknesses of the sample reaching the asymptotic value were different depending on material properties (Gogonin, 2009; Golobič and Bergles, 1997; Tachibana et al., 1967). Tachibana et al. (1967) performed a series of pool boiling experiments with various metal flat plates and then proposed a parameter, the heat capacity per unit area (the product of the volumetric heat capacity and the thickness), to describe their experimental results. Saylor (1989) proposed a parameter called thermal activity (the product of the heater thickness and the thermal effusivity) to correlate the heater property effects on CHF. Golobič and Bergles (1997) evaluated several thermal properties including the thermal activity based on their extensive experimental data, and concluded that the thermal activity best describes experimental data compared to other surface parameters.

Based on the above review, we conducted additional experiments with different thicknesses of ZIRLO and 304 stainless steel samples to explore these effects on CHF and discuss them in the context of previous studies. Even though the tested samples were not ATF materials, the experiments with ZIRLO samples of thicknesses similar to cladding wall thickness offer fundamental insights of these non-hydrodynamic effects on CHF in nuclear cladding design. We also plot together the CHF result of the Cr-coated ZIRLO samples. We measured thermal properties of ZIRLO and Cr-coated ZIRLO using laser flash thermal conductivity methods (CMC Laboratories, Tempe USA), and used literature values for 304 stainless steel. For Cr-coated samples, we measured thermal properties of 100 μm Cr coated ZIRLO and non-coated ZIRLO samples, and then we estimated thermal properties of the Cr coated samples used in CHF experiments based on the assumption of linear property changes for the thin coating samples. Based on these parameters, we calculated the heat capacity per unit area and the thermal activity and then plotted these parameters against the CHF measurements. Fig. 7 shows this comparison and shows that CHF increases as both the heat capacity per unit area and the thermal activity increase. An average thickness of the Cr coated samples of 73 μm is used in boiling. Cr-coated ZIRLO deviated slightly from the trend but the small difference in CHF is attributed to the contact angle change of Cr-coated samples compared to the others. It is important to note in this figure is that the cladding thickness are lower than the threshold thickness required for reaching the asymptotic CHF value. And, even in metal alloy such as zirconium alloy and 304 stainless steel, the small difference in thermal properties could make noticeable deviation in CHF at the same thickness. Therefore, we can conclude that the thermal properties of cladding materials in LWR coolant would be considered as one of dominant parameters affecting CHF empirical values.

4. Conclusions

In this study, we have examined CHF of candidate ATF cladding concept materials and discussed the potential challenges in analyzing CHF on the newly developed materials. While some previous experiments reported CHF values for some ATF candidate materials, the results are highly scattered depending on the coating methods and parameters that were used. In this study, we used industry scale cold spray coating technology with optimized parameters developed specifically for LWR cladding and evaluated CHF. We found that the surface parameters associated with hydrodynamics of boiling are not changed to any significant degree before and after the coating process regardless of the coating materials due to the post-surface polishing treatment required to bring it to the surface finish that is prototypical in LWR cladding. All samples show almost similar CHF values as expected from the surface characteristics. For a more in-depth understanding, we also

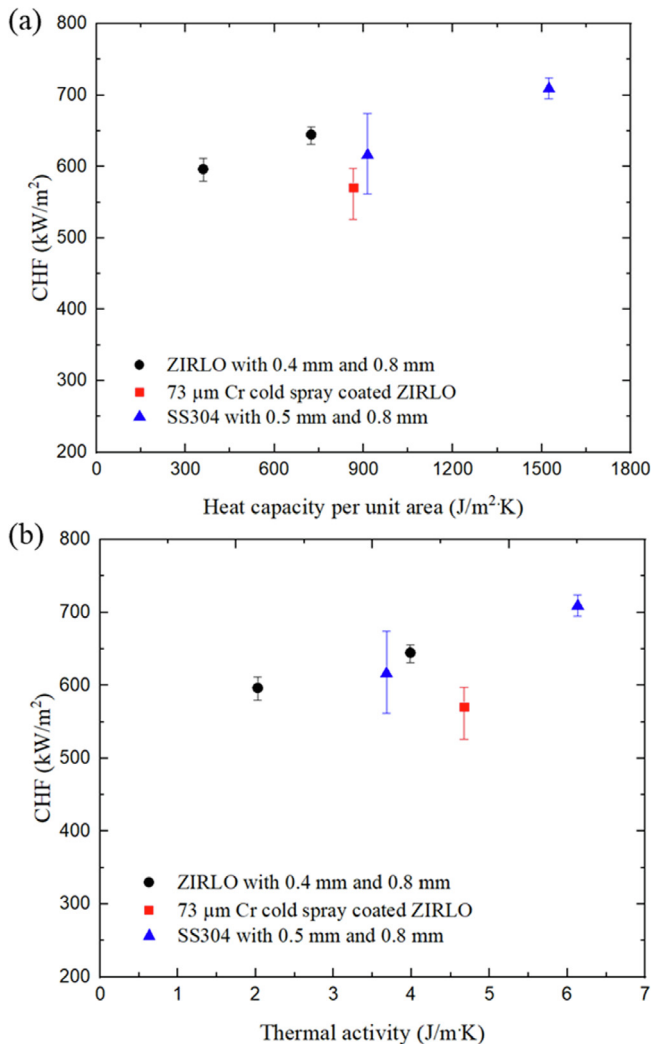


Fig. 7. CHF as a function of non-hydrodynamic parameters: (a) heat capacity per unit area [$0.5 \times \text{thickness} \times \text{heat capacity} \times \text{density}$] and (b) thermal activity [$\text{thickness} \times (\text{thermal conductivity} \times \text{heat capacity} \times \text{density})^{0.5}$].

evaluated the effect of non-hydrodynamic parameters on CHF. With the review regarding the thermal properties and thickness effects, we conducted additional experiments to understand how metal alloys with thickness similar to actual cladding could affect CHF. We found that the sub-millimeter cladding thickness could influence CHF, which needs to be incorporated to understand CHF of the new ATF cladding materials. In future studies, we will extend boiling condition from pool boiling to flow boiling to evaluate CHF of ATF candidate materials under prototypical conditions of nuclear reactor.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Acknowledgements

The authors are grateful to Westinghouse Electric Company for providing the ZIRLO materials. This work is supported by U.S. Department of Energy, Nuclear Energy University Program (NEUP)

grant DE-NE0008681.

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